Deep Learning for medical Imaging

Classification on BUSI dataset

In this project, the BUSI (Breast Ultrasound Images) dataset was used to classify images into three categories:

- Normal (N)
- Benign (B)
- Malignant (M)

Initial class distribution

```
class
benign       437
malignant    210
normal       133
Name: count, dtype: int64
```

Methodology

First, the dataset was organized by collecting all ultrasound images and assigning their corresponding labels. The mask images were removed, and only the original images were used for training the model.

To avoid data leakage, the dataset was divided into three parts:

- Training data – 70%
- Validation data – 15%
- Testing data – 15%

```
Found 546 validated image filenames belonging to 3 classes.
Found 117 validated image filenames belonging to 3 classes.
Found 117 validated image filenames belonging to 3 classes.
```

All images were resized to **224 × 224 pixels**, which is the required input size for the model. The pixel values were also normalized by dividing by **255** so that the values lie between **0 and 1**.

To increase the diversity of training data and reduce overfitting, data augmentation was used. These include rotation, zoom, width shift, height shift and horizontal flip. After applying data augmentation, random oversampling was used to balance the dataset and reduce class imbalance.

For classification, a deep learning model based on **VGG16** was used. VGG16 is a pretrained convolutional neural network. The convolutional layers were used as a feature extractor, and additional layers such as Dense and Dropout layers were added for classification into the three classes.

The model was trained using the Adam optimizer and categorical cross-entropy loss function.

Results

After training the model, the performance was evaluated using the testing dataset, which contains images that were not used during training.

The model predictions were analyzed using the following evaluation metrics:

- Precision
- Recall
- F1-score

Result for data augmentation:

```
pred = model.predict(test_data)
pred_classes = np.argmax(pred,axis=1)
true_classes = test_data.classes
print(classification_report(true_classes,pred_classes))
```

```
8/8 ━━━━━━━━━━━━━━━━━━━━ 3s 314ms/step
              precision    recall  f1-score   support

           0       0.75      0.73      0.74        66
           1       0.65      0.65      0.65        31
           2       0.64      0.70      0.67        20

    accuracy                           0.70       117
   macro avg       0.68      0.69      0.68       117
weighted avg       0.70      0.70      0.70       117
```

Result for random over sampling

Training set before oversampling

```
class
benign       306
malignant    147
normal        93
Name: count, dtype: int64
```

Training set after oversampling

```
class
benign       306
normal       306
malignant    306
Name: count, dtype: int64
```

```
train_data_over = train_gen.flow_from_dataframe(train_df_over,x_col="filename",
                  target_size=(224,224), batch_size=16,class_mode="categorical"
```

```
Found 918 validated image filenames belonging to 3 classes.
```

```
8/8 ━━━━━━━━━━━━━━━━━━━━ 2s 191ms/step
              precision    recall  f1-score   support

           0       0.77      0.67      0.72        66
           1       0.60      0.68      0.64        31
           2       0.64      0.80      0.71        20

    accuracy                           0.69       117
   macro avg       0.67      0.71      0.69       117
weighted avg       0.70      0.69      0.69       117
```